T-T-881D
February 12, 1969
SUPERSEDING
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October 8, 1965

FEDERAL SPECIFICATION

TWINE, COTTON, SEINE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements of cotton seine twine.

1.2 Classification.

1.2.1 Types and sizes. The twine shall be of the following types and in the sizes specified in table I (see 6.2):

Type I - Natural.
Type II - Mildew resistant.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

CCC-T-191 - Textile Test Methods.
PPP-B-636 - Box, Fiberboard.

Federal Standard :

Fed. Std. No. 123 - Marking for Domestics Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.)

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-C-3131 - Cordage; Preparation for Delivery of.
MIL-T-3530 - Thread and Twine; Mildew Resistant or Water Repellent Treated.

Military Standard :

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4020

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3. REQUIREMENTS

3.1 The requirements specified in 3.7 apply only to twine purchased directly by the Government. All other requirements apply to twine purchased by a supplier as a component for an end item and to twine purchased directly by the Government.

3.2 Material. The twine shall be fabricated from cotton of suitable grade and staple to meet the requirements of this specification.

3.3 Construction and physical requirements. The twine shall be of 3-strand twisted construction (cable lay), "Z" twist, each strand composed of yarns twisted in the opposite direction and shall conform to the requirements specified in table I when tested as specified in 4.2.5.

TABLE I. Physical requirements

Size (Commercial number)	Plies per strand	Length per pound, feet (minimum)	Breaking strength pounds (minimum)
6	2	3,000	12
9	3	2,055	18
12	4	1,545	24
15	5	1,245	30
18	6	1,020	35
24	8	780	45
30	10	600	60
36	12	510	70
48	16	375	86
60	20	280	108
72	24	255	124
84	28	210	140
96	32	195	156
108	36	165	172
120	40	150	188
144	48	135	214
168	56	105	238

3.4 Color. Unless otherwise specified (see 6.2), the twine shall be furnished in the natural color. When mildew resistant twine is specified, the color of the twine shall be natural prior to the mildew-resistant treatment and the shade imparted by the treatment will be acceptable.

3.5 Nonfibrous material. The starch and protein content including chloroform-soluble and water-soluble material for untreated twine shall not exceed 5.0 percent when tested as specified in 4.2.5.

3.6 Finish.

3.6.1 Type I (natural finish). Type I twine shall have a natural finish and shall conform to all the requirements specified herein. Natural finish shall denote a twine for which no consistency or lustre, other than that inherent with the cotton material, is required.

3.6.2 Type II (mildew-resistant treated). Type II twine shall conform to the requirements specified for type I and in addition shall be mildew-resistant treated in accordance with type I, class 1 treatment of MIL-T-3530.

3.7 Put-up.

3.7.1 The twine shall be furnished in 2, 4, 8 and 16-ounce minimum net weight balls or tubes as specified (see 6.2). A plus or minus tolerance of 10 percent in weight shall be allowed on each ball or tube provided the average net weight does not fall below the specified minimum when examined as specified in 4.2.3.

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3.7.2 When specified (see 6.2), the twine shall be put up on tubes and shall contain a minimum of 210 feet to a maximum 230 feet in length.

3.7.3 Each ball or tube of twine shall contain not more than the applicable maximum allowable number of knots and splices specified in table II and no piece shall be less than 25 feet in length. The twine shall be continuous throughout and shall be so wound that each turn and layer is free from entanglement.

TABLE II. Maximum allowable number of knots and splices 1/

Put-up	Maximum allowable number of knots and splices per ball or tube
2-ounce ball or tube, and 210-feet length per tube	0
4- and 8- ounce ball or tube	1
16- ounce ball or tube	2

1/ A knot shall be considered to exist when all 3 strands are separated and tied in one place joining two distinct pieces of twine. The term "knot" shall not include the tying of knots joining individual strands of twine.

3.8 Workmanship. The twine shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQL s).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item for visual defects. The defects listed in table III shall be counted regardless of their proximity to each other. The sample unit for this examination shall be one ball or tube. Ten percent of the length contained on each sample unit, but not less than 100 feet, shall be subjected to the visual examination. The lot size for this examination shall be expressed in units of balls or tubes each. The acceptable quality levels shall be 1.5 major defects and 4.0 total defects (major and minor combined) per 100 units. The inspection level shall be level I.

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TABLE III. Visual examination

Examine	Defect	Classification Major	Classification Minor
Appearance and workmanship	Any cut.	X	
	Chafed or damaged.	X	
	Kink.	X	
	Broken or loose end, or ply.	X	
Cleanness	Spot or stain 1/.		X
	Lumps, picker, or similar dirty waste and linters.		X
Color	Not natural or not color specified.	X	
Finish	Finish other than specified.	X	

1/ Visible at normal inspection distance (approximately 3 feet).

4.2.3 Examination for length, weight, and winding. The sample unit for this examination shall be one tube or ball. For lots consisting of 500 or fewer units, the sample size (number of sample units) shall be 10 and the acceptance number 1. The lot size shall be the number of tubes or balls in the inspection lot. For lots consisting of 501 or more units, the inspection level shall be S-3 and the AQL shall be 4.0 percent defective. Defects shall be as listed in 4.2.3.1, 4.2.3.2, and 4.2.3.3.

4.2.3.1 Examination for weight.

4.2.3.1.1 Net weight per ball or tube. A defect with regard to weight shall be considered to exist if the net weight on any one ball or tube is less than the minimum or more than the maximum specified in 3.7.1.

4.2.3.1.2 Average weight. No AQL is applicable to the examination for average weight. A lot shall be unacceptable if the average weight is less than the applicable minimum weight specified in 3.7.1.

4.2.3.2 Examination for winding per ball or tube. Defects with regard to winding shall be considered to exist if any of the following are determined during inspection:

a. Improperly or not firmly wound resulting in kinking, entangling or slippage during unwinding, or otherwise affecting free unhampered unwinding of twine.

b. Put-up not as specified.

c. Any holder containing more than the maximum allowable number of knots and splices specified for applicable put-up, (see table II).

d. Any knotted piece less than 25 feet in length.

e. Not in a continuous length.

4.2.3.3 Examination for length (when applicable). Defects with regard to length shall be considered to exist if the length per tube is less than the minimum or more than the maximum specified in 3.7.2 when measured in a relaxed condition.

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4.2.4 Examination of preparation for delivery requirements. An examination shall be made to determine that the packaging, packing and marking requirements of section 5 of this specification are complied with. The examination shall be in accordance with the provisions of MIL-C-3131, except that the inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

4.2.4.1 Examination for palletization. An examination shall be made to determine that the palletization requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

Examine	Defect
Finished dimension	Length, width or height exceeds specified maximum requirement.
Palletization	Not as specified. Container pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

4.2.5 Testing of the end item. The methods of testing specified in CCC-T-191, wherever applicable and as listed in table V shall be followed. When the data in the "Number of determinations" and "Results reported as" columns are not specified in the table, they shall be as required by the referenced test method. The physical and chemical values specified in section 3 apply to the average of the determination made on a sample unit for test purposes as specified in the applicable test methods. The sample size shall be in accordance with table IV. The sample unit for testing shall be one ball or tube of twine. The lot size shall be expressed in units of one ball or tube. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more units fail to meet any requirement. Tests to determine compliance with specification requirements including quantity of delivery may be made under prevailing atmospheric conditions, except in settlement of dispute in which case the test shall be made upon material which has reached equilibrium under standard conditions as defined in CCC-T-191.

TABLE IV. Sampling for tests

Lot size (balls or tubes)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE V. Test methods

Characteristic	Requirement paragraph	Test method	Number of determinations per individual sample unit	Results reported as:
Material identification	3.2	1200 1/	--	-
Number of strands	3.3	Visual	1	Pass or fail
Direction of twist (Strands and twine)	3.3	4050	3	Pass or fail
Number of plies per strand	Table I	Visual	1	Pass or fail
Length per pound	Table I	4010 2/	2	-
Breaking strength	Table I	4102	5	-
Nonfibrous material	3.5	2611	-	-

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Except that the test specimen shall be a minimum of 30 yards of twine and the results shall be converted to feet per pound.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Twine, put-up as specified, shall be packaged in accordance with the applicable require of MIL-C-3131.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B or C. Except when otherwise specified by civil agencies, twine shall be packed in accordance with the applicable requirements of MIL-C-3131.

5.2.1.1 Standard pack (civil agencies). One hundred and twenty 1/2 pound balls or tubes, or sixty one pound balls or tubes, of one type and size (commercial number) only shall be packed in a close-fitting box conforming to PPP-B-636, class domestic, style CSSC, FOL or SFF at the option of the supplier. The fiberboard shall meet the special requirements of table I of PPP-B-636. The box shall be closed in accordance with the appendix of the box specification.

5.3 Palletization. Unless otherwise specified (see 6.2), twine packed as specified in 5.2.1, shall be palletized in accordance with the applicable requirements of MIL-C-3131. The load type shall be as specified by the procuring activity (see 6.2).

5.4 Marking.

5.4.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military agencies. In addition to markings required by the contract or order, the interior packages, shipping containers and palletized unit loads shall be marked in accordance with MIL-C-3131.

6. NOTES

6.1 Intended use. The twine covered by this specification is intended for general purpose use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Size (Commercial number) required (see table I).
(d) Color of twine if other than natural (see 3.4).

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(e) Put-up required (see 3.7.1).
(f) When 210-foot tubes are required (see 3.7.2).
(g) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(h) When palletization is not required (see 5.3).
(i) Selection of applicable load types for palletization, when required (see 5.3).

6.3 Standard pack for civil agencies. The standard pack requirements in 5.2.1.1 are intended for use in procurement of stores stock replenishments. Procuring officers should use the standard pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or intermediate storage point for temporary storage, subsequent issue, or shipment to eventual users.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 82

Review activities:

Army - GL, MD
Navy - SA
Air Force - 82, 85

User activities:

Army - ME
Navy - SH, MC

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

AGR
COM-NBS
DC
GSA-FSS
HEW
INT
JUS-FPI
PO
TR
VA-DMS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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